

2. The total stopping distance for a moving car is given by the equation below:

$$\text{total stopping distance} = \text{thinking distance} + \text{braking distance}$$

- (a) These distances may be affected by a number of factors.
Three of these factors are given in the table below.

Put a **tick** (✓) or a **cross** (X) in each box below to show whether the distance is affected by each factor. [3]

The first row has been done for you.

Factor	Thinking distance	Braking distance	Total stopping distance
Worn tyres	X	✓	✓
Drunk driver			
Wet road			

- (b) At a speed of 13 m/s, the thinking distance of an alert driver is 9.1 m.

Use the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

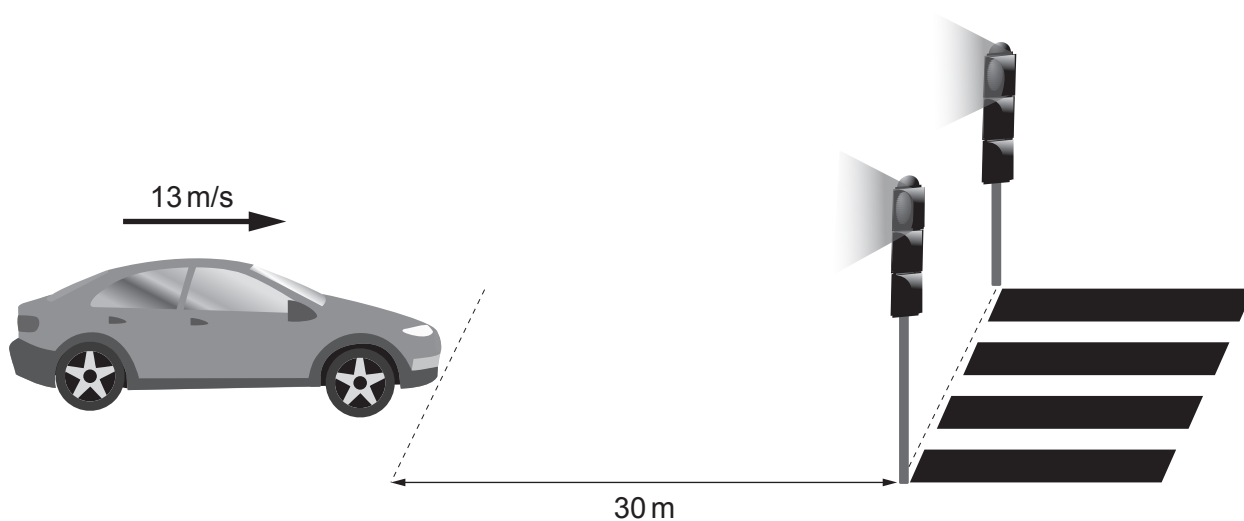
to calculate the thinking time.

[2]

Thinking time = s



- (c) A driver of a car travelling at 13 m/s sees traffic lights 30 m ahead when the lights turn to red.



The thinking distance = 9.1 m and the braking distance = 13.9 m at this speed.

Use the equation:

total stopping distance = thinking distance + braking distance

to explain whether the car would be able to stop before reaching the crossing. [2]

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